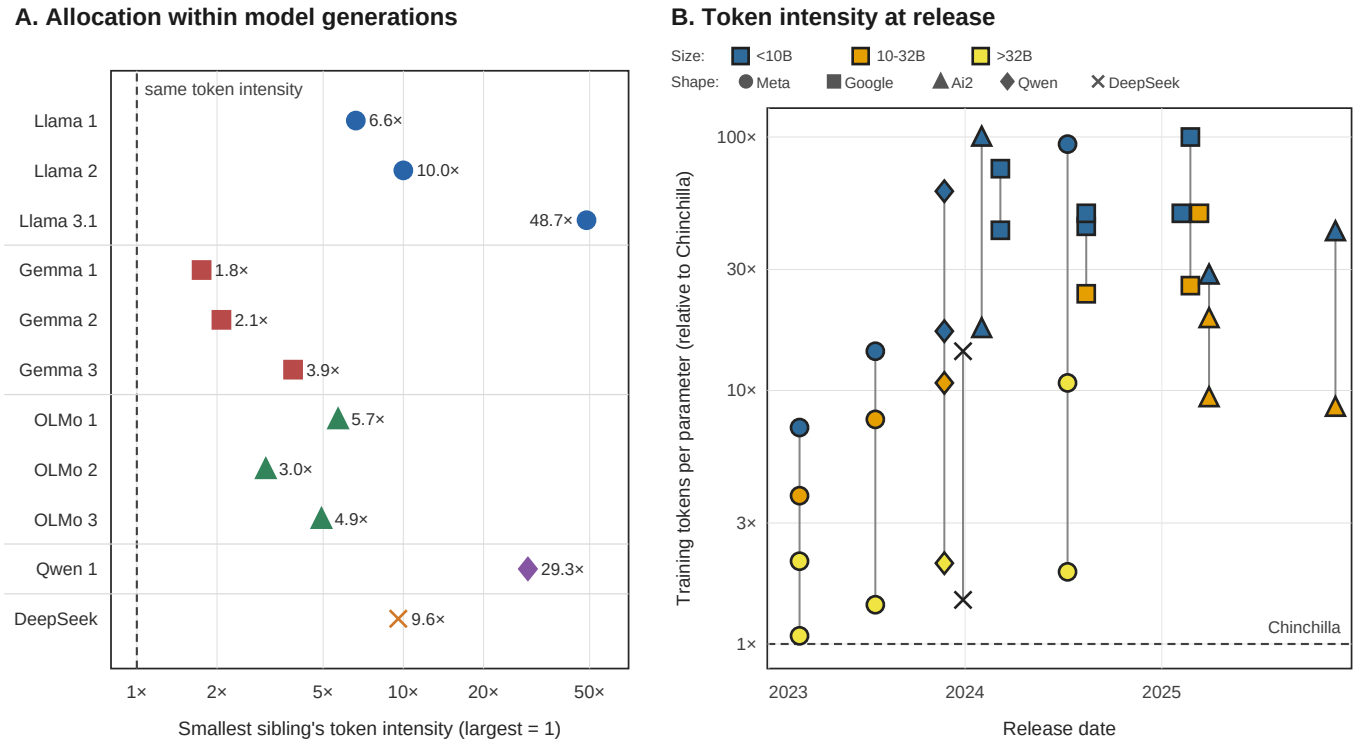
A. Allocation within model generations
same token intensity
Llama 1
Llama 2
Llama 3.1
Gemma 1
Gemma 2
Gemma 3
OLMo 1
OLMo 2
OLMo 3
Qwen 1
DeepSeek
6.6×
10.0×
48.7×
1.8×
2.1×
3.9×
5.7×
3.0×
4.9×
29.3×
9.6×
1×
2×
5×
10×
20×
50×
Smallest sibling's token intensity (largest = 1)
B. Token intensity at release
Size: <10B 10-32B >32B
Shape: Meta Google Ai2 Qwen DeepSeek
Training tokens per parameter (relative to Chinchilla)
100×
30×
10×
3×
1×
Chinchilla
2023
2024
2025
Release date